

Customer Churn Prediction

Model Results Report

Teyzix Core Internship (June Batch)
Task DA-INT-1 • Domain: Data Analytics

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1. Executive Summary

This report documents the development and evaluation of machine-learning models for predicting customer churn on the Telco Customer Churn dataset (7,043 customers, 21 attributes). Three classifiers were trained and compared: Logistic Regression, Random Forest, and XGBoost. After comparing performance across five metrics, Logistic Regression was selected as the production model.

Selection rationale: Logistic Regression delivered the highest churn-class recall (0.783) and the highest ROC-AUC (0.841) of the three models. Because the business objective is to identify as many at-risk customers as possible before they leave, recall and ranking quality (AUC) are the most relevant criteria. The trade-off is a lower churn-class precision (0.507 versus 0.549 for Random Forest), meaning more false alarms — an acceptable cost when a missed churner is more expensive than an unnecessary retention offer.

2. Methodology

2.1 Data Preparation

- TotalCharges was converted from text to numeric; 11 blank values (all tenure = 0, i.e. brand-new customers) were set to 0.
- The customerID column was dropped as a non-predictive identifier.
- High TotalCharges values were retained, not removed: they represent genuine long-tenure customers rather than data errors.
- The target Churn was encoded as 1 (Yes) / 0 (No). Class balance is approximately 26.5% churn — a moderately imbalanced problem.

2.2 Feature Engineering

- tenure_group: tenure binned into 0–1yr, 1–2yr, 2–4yr, 4–5yr, 5yr+.
- num_services: count of subscribed add-on services (engagement proxy).
- charge_ratio and avg_monthly_total: spend pattern relative to lifetime.
- auto_payment and is_monthly: payment-method and contract-type flags.

Note on dataset limitations: the task brief lists “usage frequency,” “support interaction count,” and “trend-based features.” These cannot be engineered because the dataset contains no usage logs, support-ticket records, or timestamps. Only features genuinely derivable from the data were created.

2.3 Training Setup

- 80/20 train-test split, stratified on the target to preserve class balance.
- Features standardised for Logistic Regression; tree models used unscaled inputs.
- class_weight = 'balanced' applied to counteract the 26.5% / 73.5% imbalance.

3. Model Comparison

All three models were evaluated on the held-out 20% test set (1,409 customers) using identical preprocessing. The strongest value in each comparative row relevant to the selection is shown in bold.

Metric	Logistic Regression	Random Forest	XGBoost
Accuracy	0.740	0.769	0.770
Precision (churn)	0.507	0.549	0.551
Recall (churn)	0.783	0.730	0.727
F1 (churn)	0.616	0.627	0.627
ROC-AUC	0.841	0.840	0.827

Reading the table: the tree models (Random Forest, XGBoost) edge ahead on raw accuracy and precision, but Logistic Regression leads on recall and AUC. Since accuracy is misleading on an imbalanced dataset — a model could score 73.5% by predicting “no churn” for everyone — it was not used as the deciding metric.

3.1 Why Not Accuracy?

With only 26.5% of customers churning, accuracy rewards correctly classifying the majority “no churn” class. A churn-detection system that maximises accuracy at the expense of recall would miss the very customers it exists to catch. Recall and ROC-AUC therefore drove the decision, both of which favour Logistic Regression.

4. Selected Model — Logistic Regression

4.1 Confusion Matrix (Test Set, n = 1,409)

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	750 (True Negative)	285 (False Positive)
Actual: Churn	81 (False Negative)	293 (True Positive)

Of 374 actual churners in the test set, the model correctly identifies 293 and misses only 81 — the source of its 0.783 recall. The cost of this sensitivity is 285 false positives: non-churning customers flagged as at-risk. In practice these false alarms mean some retention spend reaches customers who would have stayed, which is generally cheaper than losing a customer who was never contacted.

4.2 Performance Metrics

- Accuracy: 0.740
- Precision (churn class): 0.507
- Recall (churn class): 0.783
- F1 (churn class): 0.616
- ROC-AUC: 0.841

4.3 Risk-Category Distribution

Predicted churn probabilities were bucketed into three risk tiers. On the test set the distribution was:

Risk Category	Probability Range	Customers (test set)
Low risk	0.00 – 0.33	629
Medium risk	0.33 – 0.66	352
High risk	0.66 – 1.00	428

5. Limitations & Recommendations

5.1 Limitations

- The dataset is a single cross-sectional snapshot with no timestamps, so genuine time-series, trend, or “weekly pipeline” analysis is not possible.
- Behavioural features named in the brief (usage frequency, support interactions) are absent from the source data and were not fabricated.
- Logistic Regression's lower precision (0.507) produces a meaningful false-positive rate; retention budgets should account for outreach to some non-churners.
- Coefficient interpretation requires care: tenure and TotalCharges are correlated, so their coefficients partially offset one another.

5.2 Recommendations

- Deploy Logistic Regression for ranking customers by churn probability, prioritising the High-risk tier for retention outreach.
- Consider tuning the decision threshold (rather than the default 0.5) to balance recall against retention-campaign cost.
- Enrich future data collection with usage and support-interaction logs to enable the behavioural features the brief envisioned.
- Re-train periodically as new churn outcomes accumulate.

6. Conclusion

Three models were trained and rigorously compared. Logistic Regression was selected as the production model on the strength of its churn-class recall (0.783) and ROC-AUC (0.841), the two metrics most aligned with the business goal of early churn detection. The model identifies roughly four out of five churners on unseen data, and its probability outputs support a practical three-tier risk-segmentation system for targeting retention efforts.